

Introduction to Quarto: Creating Homework Reports

Dr. Samuel B Fernandes

2026-01-01

Table of contents

1	What is Quarto?	2
2	Creating Your First Homework Report	3
2.1	Quick Start (VS Code)	3
2.2	Quick Start (Posit Cloud)	3
2.3	Edit in Visual Mode (VS Code)	3
2.4	Basic Homework Template	4
3	Problem 1: Calculate Yield Statistics	4
3.1	Question	4
3.2	Solution	4
4	Problem 2: Create a Summary Table	5
4.1	Question	5
4.2	Solution	5
5	Document Structure for Homework	6
5.1	YAML Header (Top of File)	6
5.2	Sections (Using Headings)	6
5.3	Text: Explanations and Interpretations	6
6	Basic Markdown for Writing Text	7
6.1	Headings	7
6.2	Text Formatting	7
6.3	Lists	7
6.4	Equations (Mathematical Notation)	7
7	Working with Code Chunks	8
7.1	What is a Code Chunk?	8
7.2	Basic Code Chunk Example	8
7.3	Adding Labels and Captions	8
7.4	Load Packages at the Start	9

8	Best Practices for Homework	9
8.1	Keep It Organized	9
8.2	Write for the Grader	10
8.3	Common Mistakes to Avoid	10
8.4	Reproducibility Tip	10
9	Detailed Example: Complete Homework Assignment	10
9.1	Template Structure:	11
9.2	Exporting: HTML → Print to PDF (Nicer PDF)	12
10	Troubleshooting	14
10.1	Document won't render (PDF doesn't appear)	14
10.2	"Package not found" error	15
10.3	Code runs fine in R but fails in Quarto	15
10.4	Code chunk doesn't run	15
10.5	Figure doesn't appear in PDF	15
10.6	File size is too large	15
11	Resources	15
11.1	Official Quarto Documentation	15
11.2	Help & Support	16
11.3	For Posit Cloud Users	16
11.4	For VS Code Users	16

1 What is Quarto?

Quarto combines **text**, **code**, and **output** in one document that renders to PDF, HTML, Word, and more. For homework, this means you can write your analysis in one file and produce a professional PDF ready to submit.

Why use Quarto for homework?

- Write explanations and R code in the same document
- Figures and tables appear automatically in your report
- Your work is reproducible—anyone can run your code and get the same results
- No copying/pasting results between programs
- One .qmd file is easier to organize than multiple documents

File extension: .qmd (like .docx or .R)

Three main components:

1. **YAML header** – metadata (title, author, date, output format)
2. **Markdown text** – your written explanations
3. **Code chunks** – R code that runs and produces output

2 Creating Your First Homework Report

2.1 Quick Start (VS Code)

1. **Create a new file:** File → New Text File (or Ctrl/Cmd + N)
2. **Save it:** Ctrl/Cmd + S and name it `homework01.qmd`
3. **Copy the template below** into your file
4. **Render to PDF:** Click the **Render** button (top-right) or press Ctrl/Cmd + Shift + K
5. **Open the PDF:** Look for `homework01.pdf` in the same folder

Tip: VS Code uses Source mode only. Type everything as shown in the examples below.

2.2 Quick Start (Posit Cloud)

1. **Create a new file:** File → New File → Quarto Document
2. **Save it:** Click the disk icon and name it `homework01.qmd`
3. **Switch to Source mode:** Click **Source** button (top-left of editor)
4. **Copy the template below** into your file
5. **Render to PDF:** Click the **Render** button
6. **Download the PDF:** Right-click the rendered PDF and download, or check the Files pane

Visual vs. Source mode: - **Source mode** (recommended for homework): You see the raw Quarto syntax—headings with `#`, code chunks with triple backticks - **Visual mode:** WYSIWYG editor (like Word)—use the **Insert** menu to add code chunks

For this course, use Source mode so you can copy-paste examples directly.

2.3 Edit in Visual Mode (VS Code)

Prefer Source mode for this course, but if you want to try Visual editing in VS Code:

- Install the Quarto extension (Extensions → search “Quarto”).
- Open your `.qmd` file, then use the Command Palette: “Quarto: Open Document in Visual Editor” (or the toolbar button “Edit in Visual Mode” if available).
- Insert a code chunk via the Insert menu → Code Chunk → R.
- To return to Source mode, click the “Source” button at the top of the editor or use the Command Palette: “View: Toggle Editor Mode”.

Visual mode is helpful for formatting; always verify chunks look exactly like ````{r}` when you switch back to Source.

2.4 Basic Homework Template

Copy and paste this structure into your new .qmd file and customize it.

YAML Header (top of file):

```
---
title: "Homework 01: R Basics"
author: "Your Name"
date: "January 15, 2026"
format:
  pdf:
    toc: false
    number-sections: true
editor: source
---
```

Then add your problems below using markdown headings and code chunks.

3 Problem 1: Calculate Yield Statistics

3.1 Question

Calculate the mean, standard deviation, and range of the following yields (kg/ha): 2500, 2650, 2600, 2800, 2550, 2900.

3.2 Solution

First, I'll create a vector of the yields and calculate the statistics.

Now add a code chunk. **Type exactly as shown:**

```
```{r}
yields <- c(2500, 2650, 2600, 2800, 2550, 2900)
mean_yield <- mean(yields)
sd_yield <- sd(yields)
range_yield <- range(yields)
Display results
cat("Mean yield:", mean_yield, "kg/ha\n")
cat("Standard deviation:", sd_yield, "kg/ha\n")
cat("Range:", range_yield[1], "to", range_yield[2], "kg/ha\n")
```
```

Then continue with markdown text to interpret the results:

****Interpretation:**** The yields vary from 2500 to 2900 kg/ha with a mean of 2675 kg/ha.

4 Problem 2: Create a Summary Table

4.1 Question

Create a data frame with treatment names and their corresponding yields, then calculate the mean yield by treatment.

4.2 Solution

Add another code chunk. **Type exactly as shown:**

```
```{r}
library(dplyr)

Create the data
trial <- data.frame(
 treatment = c("Control", "N50", "N100", "N150"),
 yield = c(2400, 2650, 2900, 2950)
)

Calculate mean yield
mean_yield <- mean(trial$yield)

cat("Overall mean yield:", mean_yield, "kg/ha\n")
cat("\nTreatment yields:\n")
print(trial)
```
```

Then add interpretation text:

The N150 treatment produced the highest yield at 2950 kg/ha.

How to create code chunks:

1. Type three backticks: ` ` `
2. Immediately type {r}
3. Press Enter and type your R code
4. Type three more backticks to close: ` ` `

The backticks wrap your R code so Quarto knows to execute it.

Save this template, change the title and problems to match your assignment, and fill in your solutions.

5 Document Structure for Homework

5.1 YAML Header (Top of File)

The YAML header contains metadata about your document:

```
---
title: "Homework 01: R Basics"
author: "Your Name"
date: "January 15, 2026"
format:
  pdf:
    number-sections: true
editor: source
---
```

What each option does:

- `title` – appears at the top of the PDF
- `author` – your name
- `date` – when you completed it
- `format: pdf:` – renders to PDF (what you submit)
- `number-sections: true` – numbers your sections (1, 2, 3...)

5.2 Sections (Using Headings)

Use `#` for main sections and `##` for subsections:

```
# Problem 1: Data Exploration

## Question
What is the mean yield?

## Solution
[Your code and answer here]
```

This creates a neat structure in your PDF.

5.3 Text: Explanations and Interpretations

Write plain text to explain your thinking:

```
I calculated the mean by adding all yields and dividing by the number of observations.
The result shows that nitrogen fertilizer increases yield by approximately 200 kg/ha.
```

6 Basic Markdown for Writing Text

6.1 Headings

```
# Main Section (Problem 1)
## Subsection (Question)
### Sub-subsection
```

6.2 Text Formatting

- **`**bold text**`** makes text bold
- *`*italic text*`* makes text italic
- ``code`` displays code inline: `mean(x)`

6.3 Lists

Bullet points:

- First item
- Second item
- Third item

Numbered lists:

1. First step
2. Second step
3. Third step

6.4 Equations (Mathematical Notation)

For simple math inline, use dollar signs: $\mu = \bar{x}$

For a centered equation:

```
$$y = \beta_0 + \beta_1 x + \epsilon$$
```

This renders nicely in your PDF. (Don't worry about advanced LaTeX—stick to simple notation for homework.)

7 Working with Code Chunks

7.1 What is a Code Chunk?

A code chunk is R code that runs when you render the document.

To create a code chunk:

1. Type three backticks: `` ` ``
2. Immediately add `{r}` with no space
3. Press Enter and write your R code
4. Close with three more backticks: `` ` ``

Example of what you type:

```
```{r}
Your R code here
```
```

Visual example:

Type ````{r}` then your code, then `````

In Visual mode (Posit Cloud): Use Insert → Code Chunk → R, or press Ctrl+Alt+I (Windows) or Cmd+Option+I (Mac).

7.2 Basic Code Chunk Example

Here's what you type in your `.qmd` file:

```
```{r}
Calculate basic statistics
yields <- c(2500, 2650, 2600, 2800, 2550, 2900)
mean(yields)
```
```

When you render, this code runs and the result (the mean) appears in your PDF.

Common mistake: Forgetting the `{r}` part. It must be ````{r}` not just `````.

7.3 Adding Labels and Captions

For figures, add a caption using special `#|` comments at the top of the chunk.

Example chunk options for figures:

- `#| fig-cap: "Your caption text"` – adds a caption
- `#| fig-width: 6` – sets width in inches
- `#| fig-height: 4` – sets height in inches

- `#| label: fig-myplot` – gives the figure a label for cross-referencing

Example:

```
```{r}
#| fig-cap: "Yield response to nitrogen fertilizer"
#| fig-width: 6
#| fig-height: 4

library(ggplot2)
data <- data.frame(
 treatment = c("Control", "N50", "N100", "N150"),
 yield = c(2400, 2650, 2900, 2950)
)
ggplot(data, aes(x = treatment, y = yield)) +
 geom_col(fill = "steelblue") +
 labs(x = "Treatment", y = "Yield (kg/ha)")
```
```

Other useful options: - `echo: false` – hide code, show only output - `eval: false` – show code but don't run it

7.4 Load Packages at the Start

Create one code chunk at the beginning to load all packages:

```
```{r}
#| message: false
#| warning: false

library(dplyr)
library(ggplot2)
```
```

The `message: false` and `warning: false` options hide package startup messages in your PDF.

8 Best Practices for Homework

8.1 Keep It Organized

1. Start with a clear YAML header (title, author, date)
2. Use problem headings (`# Problem 1`, `# Problem 2`, etc.)
3. Within each problem, use subsections for Question and Solution
4. Write brief explanations before and after code

8.2 Write for the Grader

- **Explain your thinking:** Don't just show code; write a sentence about what you're doing
- **Interpret results:** If you calculate a mean, say what it means
- **Show all work:** Include the R code that produces your answer
- **Answer the question directly:** Make it clear you understood what was asked

Tip

Start simple: For each problem, write one sentence explaining what you'll do, a small code chunk that runs, and one sentence interpreting the result. If stuck, paste the copy-all template above and change only the titles, numbers, and names.

8.3 Common Mistakes to Avoid

Missing packages: If you use `ggplot2`, make sure you run `library(ggplot2)` first

Broken code: Test your code in R before submitting—if it doesn't run for you, it won't run for the grader

No output: Make sure your code chunk actually produces output (the answer) in the PDF

Unclear explanations: Don't assume the grader knows what you're doing—explain your steps

8.4 Reproducibility Tip

Use `set.seed()` if your code involves randomness:

```
```{r}
set.seed(42) # Everyone gets the same random numbers
data <- rnorm(100, mean = 100, sd = 15)
```
```

This ensures your results are identical every time you render.

9 Detailed Example: Complete Homework Assignment

Below is a full homework example showing all the pieces together. **This is what your complete .qmd file should look like.**

Copy this entire template structure and modify for your homework:

9.1 Template Structure:

Copy the entire block below into a new file named `homework01.qmd`, then update the title, author, and problems. This single chunk includes everything you need.

```
# Copy-All Homework QMD Template

---
title: "Homework 01: Data Exploration with R"
author: "Your Name"
date: "January 15, 2026"
format:
  pdf:
    number-sections: true
editor: source
---

# Problem 1: Calculate Yield Statistics

## Question
Write the question text here.

## Solution
Write your explanation, then add a code chunk.

```{r}
library(dplyr)
yields <- c(2500, 2650, 2600, 2800, 2550, 2900)
mean(yields)
```

Briefly interpret the result in one sentence.

# Problem 2: Data Frame Analysis

## Question
Another question here.

## Solution
Explanation, then code, then interpretation.

```{r}
trial <- data.frame(
 treatment = c("Control", "N50", "N100", "N150"),
 yield = c(2400, 2650, 2900, 2950)
)
mean_yield <- mean(trial$yield)
```

```
cat("Overall mean yield:", mean_yield, "kg/ha\n")
```
```

```
# Problem 3: Create Visualization
```

```
## Question
Create a plot...
```

```
## Solution
Use ggplot2 to create your figure.
```

```
```{r}
#| fig-cap: "Yield by treatment"
#| fig-width: 6
#| fig-height: 4
library(ggplot2)
data <- data.frame(
 treatment = c("Control", "N50", "N100", "N150"),
 yield = c(2400, 2650, 2900, 2950)
)
ggplot(data, aes(x = treatment, y = yield)) +
 geom_col(fill = "steelblue") +
 labs(x = "Treatment", y = "Yield (kg/ha)")
```
```

```
# Package Loading (Optional at Top)
```

```
```{r}
#| message: false
#| warning: false
library(dplyr)
library(ggplot2)
```
```

```
# Reproducibility Tip
```

```
```{r}
set.seed(42)
```
```

After pasting, change the title/date and customize the Problem sections.

9.2 Exporting: HTML → Print to PDF (Nicer PDF)

Sometimes HTML printed to PDF looks nicer than direct PDF:

- Render HTML: Run from terminal: `quarto render homework01.qmd --to html`, or set `format: html` in the YAML and click Render.
- Open the HTML in your browser.
- Use the browser's Print → Save as PDF.
- If figures look too large, adjust `#| fig-width` and `#| fig-height` in chunks and re-render.

YAML Header (at the very top):

```
---
title: "Homework 01: Data Exploration with R"
author: "Your Name"
date: "January 15, 2026"
format:
  pdf:
    number-sections: true
editor: source
---
```

Problem 1 Section:

Use a level-1 heading for the problem number, then Question and Solution subsections:

```
# Problem 1: Calculate Yield Statistics
```

```
## Question
```

Write the question text here.

```
## Solution
```

Write your explanation, then add a code chunk.

Then add your R code chunk (three backticks, then `{r}`):

```
library(dplyr)
yields <- c(2500, 2650, 2600, 2800, 2550, 2900)
mean(yields)
```

Close with three more backticks, then interpret the results.

Problem 2 Section:

Follow the same pattern:

```
# Problem 2: Data Frame Analysis
```

```
## Question
```

Another question here.

Solution

Explanation, then code, then interpretation.

Problem 3 Section:

Include visualization:

Problem 3: Create Visualization

Question

Create a plot...

Solution

Use ggplot2 to create your figure.

Important: - Each problem gets its own level-1 heading (#) - Use ## for Question and Solution subsections - Always explain before and after your code - Make sure all code actually runs before submitting

This template ensures: - Clear problem statement - Clean code that runs without errors - Results displayed in the PDF - Written interpretations explaining what the results mean - Professional organization

10 Troubleshooting

10.1 Document won't render (PDF doesn't appear)

Check these in order:

1. **Save your file:** Make sure you saved as `.qmd` (not `.txt`)
2. **Check for errors:** Look at the rendering messages in the console. Copy-paste error messages into a search engine if stuck.
3. **Test the code separately:** Run your R code in a regular R script to make sure it works before putting it in Quarto
4. **YAML indentation:** Make sure your header uses spaces, not tabs. The title, author, date should line up.

10.2 “Package not found” error

If you get an error like “there is no package called ‘dplyr’”:

Run this in R console: `install.packages("dplyr")`

Then render again.

10.3 Code runs fine in R but fails in Quarto

Make sure you’ve loaded the package in a code chunk before using functions from it:

```
```{r}
library(ggplot2) # Load first
```
```

Then use ggplot commands in a later chunk.

10.4 Code chunk doesn’t run

Check: - Did you type ````{r}` exactly (with curly braces)? - Did you close the chunk with `````? - Are there matching backticks (3 to open, 3 to close)?

10.5 Figure doesn’t appear in PDF

- Check that your code chunk has proper `{r}` syntax
- Make sure the code actually produces a plot (no errors)
- Try adding these options: `#| fig-width: 6` and `#| fig-height: 4`

10.6 File size is too large

If your PDF is huge (>10 MB): - Reduce figure sizes with `fig-width: 5` and `fig-height: 4` - Use lower resolution: `fig-dpi: 100` instead of default - Remove unnecessary code chunks

11 Resources

11.1 Official Quarto Documentation

- **Quarto Guide:** <https://quarto.org/docs/guide/>
- **Getting Started:** <https://quarto.org/docs/get-started/>

11.2 Help & Support

- **Quarto FAQ:** <https://quarto.org/docs/faq/>
- **RStudio Community:** <https://community.rstudio.com/> (ask questions here!)
- **Stack Overflow:** Tag your questions with `quarto`

11.3 For Posit Cloud Users

If Quarto Preview fails in Posit Cloud: 1. Use the **Render** button instead of Preview 2. Check that your `.qmd` file is saved 3. Make sure your code runs without errors first

11.4 For VS Code Users

- Install the Quarto extension: Extensions → search “Quarto”
 - Install the R extension for syntax highlighting
 - Use **Render** button or Ctrl/Cmd + Shift + K
-

End of Introduction to Quarto

For questions or help, contact Dr. Samuel B Fernandes or visit office hours.